TRAINING CURRICULUM

Hasta EMS Trainer Programme

Curriculum v1.1 · whole-body EMS (WB-EMS) · for Hasta Certified EMS Trainers

Status	**DRAFT — for approval by the medical lead in the EMS app (Academy → Curriculum)**
Replaces	Curriculum v1.0 (draft, never approved)
Prepared	13 November 2026
Theory and skills	24 hours across modules 1–3 and 5–10, plus the supplier's machine course (module 4)
Supervised practice	At least 10 logged client sessions, including 3 first sessions, under a Hasta Master Trainer (module 11)
Certification	Theory test (40 questions, pass mark 80%, up to 3 attempts) + every practical checklist item passed + supervised practice signed off. Certificate is per machine model, valid 12 months, with a public verification code.
Based on	Hasta EMS Academy framework v1.0; the international guideline for safe and effective WB-EMS (Kemmler et al., *Frontiers in Physiology*, 2023); the revised German consensus on WB-EMS contraindications (von Stengel et al., *Frontiers in Sports and Active Living*, 2024); the equipment supplier's instructions.

Before this curriculum is used

Every clinical rule in this document — contraindications, first-session limits, the gap between sessions and when to refer — must be confirmed by the medical lead. Where Hasta's current rules differ from the international guideline, this is marked DECISION for the medical lead to settle before approval.

Contents

Part A	How the programme works — who it is for, entry requirements, structure, assessment and certification
Part B	The modules 1. How EMS works 2. Client screening, contraindications and consent 3. Rhabdomyolysis, preparation and recovery 4. Machine module (per model) 5. Running a session 6. Programme design and progression 7. Emergencies, adverse events and first aid 8. Measurements, progress and lifestyle support 9. Hasta protocols, the platform and data protection 10. Client care, ethics and professional conduct 11. Supervised practice
Part C	Tools — training timetable, pre-session checklist, Borg CR10 scale, conditioning-phase plan, practical assessment checklist, decisions for the medical lead

Part A — How the programme works

Who it is for

Staff who will run whole-body EMS sessions with Hasta clients: wellness coaches, physiotherapists, nurses and doctors. Nobody may run an EMS session at Hasta without a current Hasta Certified EMS Trainer certificate for that machine model, except as a trainee under direct Master Trainer supervision during module 11.

Entry requirements

- A recognised fitness, physiotherapy, nursing or medical qualification (the international guideline expects an official coaching qualification plus WB-EMS-specific training).
- A current Basic Life Support (BLS) certificate, or completion before module 7 is signed off.
- An account in the Hasta EMS app with the Trainer skill.

The certification chain

Level	Who	Requirement
Level A external qualification	Master Trainer candidates	Supplier or external WB-EMS qualification covering the machine, verified in the platform by someone other than the holder.
Hasta Master Trainer	Wellness coaches or doctors	Verified Level A for the machine + Hasta Master Trainer course (renewed every 24 months). Supervises trainees and signs off practicals.
Hasta Certified EMS Trainer	All EMS trainers	This curriculum, the theory test, the practical checklist and 10 supervised sessions. Per machine model, valid 12 months; renewal needs at least 20 logged sessions in the year and a refresher test.

How it is delivered

Part	Format	Hours
Modules 1–3, 5–10	Classroom teaching, demonstrations, case studies and role-play, led by a Master Trainer; the medical lead teaches modules 2, 3 and 7 or approves the trainer who does.	24
Module 4	The equipment supplier's official course on the machine model, with Hasta's set-up and cleaning routine.	Per supplier (at least 4)
Module 11	Real client sessions, logged in the EMS app, with the Master Trainer present throughout.	10 sessions

Assessment

- **Theory test** in the EMS app: 40 questions drawn from the approved question bank — How EMS works 6, Contraindications 10, Rhabdomyolysis 6, Machine 6, Running a session 5, Emergencies 4, Protocols 3. Pass mark 80%. Up to 3 attempts; after a failed attempt the Master Trainer reviews the weak areas with the trainee before the next one.
- **Practical checklist** (Part C): 16 items observed by the Master Trainer. Items marked ★ are critical — one failure means the practical is not passed.
- **Supervised practice**: at least 10 sessions logged, including 3 with clients on their first EMS session.

Part B — The modules

MODULE 1 How EMS works

3 hours · theory test

Aim: trainees understand what EMS does to muscle, what it can and cannot achieve, and why the dose must be controlled.

By the end, the trainee can:

- Explain how electrical impulses delivered through electrodes cause muscle contraction, and how this differs from a voluntary contraction (larger and more simultaneous recruitment of muscle fibres).
- Describe the impulse settings the trainer controls — intensity per electrode area, impulse phase and rest phase, programme type — and what each changes for the client.
- Explain why whole-body EMS is a high muscle load in a short time, and why sessions are short and spaced out.
- State realistic benefits supported by research (strength, muscle mass and function, some help with back pain as part of a programme) and the limits — EMS is not a weight-loss treatment on its own and results vary.
- Recognise and correct misleading claims ("lose X kg", "burns fat while you rest", "replaces exercise").

Content: skeletal muscle and motor units; nerve stimulation by external current; the strength–intensity relationship; muscle damage and repair; evidence summary; Hasta's language for describing EMS to clients.

Activities: trainee experiences a short low-intensity demonstration; group exercise rewriting misleading marketing statements.

MODULE 2 Client screening, contraindications and consent

4 hours · theory + practical

Aim: no client receives EMS who should not, and every client is screened, informed and consented before the first session and checked again before every session.

By the end, the trainee can:

- Run the Hasta admission screening in the EMS app and act on every answer.
- Name the **absolute contraindications** (no EMS at Hasta): pacemaker, implanted defibrillator or any electronic implant; pregnancy; epilepsy and other neurological disorders; heart rhythm disorders and serious heart disease; arterial disease or circulation disorders; bleeding disorders; active cancer; acute illness, fever, infection or inflammation; abdominal or groin hernia; recent surgery in the area to be stimulated; acute effect of alcohol or drugs.
- Name the **relative contraindications** (only with written clearance from the clinician): diabetes; past cancer; implants or surgery older than 6 months; stents or bypass older than 6 months; kidney or other internal-organ disease; controlled high blood pressure; back pain without a diagnosis, slipped disc or nerve pain; fluid retention or oedema; open skin wounds, rashes or burns (no electrode over them); and any regular medicine the clinician wants to review.
- Re-check before every session: new illness, fever, new medicines, pregnancy, surgery, injury, alcohol in the last 24 hours, and how the client felt after the last session.

- Explain EMS, its risks and the client's right to stop, and take signed consent in the app; renew consent at least every 6 months.
- Stop and refer to the clinician when unsure — never decide a clinical exception alone.

Why the list is strict

The list follows the international guideline and the revised German consensus (2024), which still treat neurological disorders and arterial disease as absolute contraindications and require medical clearance for conditions such as diabetes or past cancer. The medical lead may tighten, but not loosen, the list for Hasta.

Activities: screening role-plays (including a client who leaves out a pacemaker, and a client whose phone number is shared with a relative); case cards — "go, stop or refer?".

MODULE 3 Rhabdomyolysis, preparation and recovery

3 hours · theory + scenario

Aim: trainees prevent, recognise and escalate exercise-induced muscle breakdown, the most serious known risk of WB-EMS.

By the end, the trainee can:

- Explain rhabdomyolysis: muscle cells break down and release their contents (including myoglobin and creatine kinase) into the blood, which can damage the kidneys.
- List the risk factors: high intensity in early sessions, too little recovery, dehydration, training while ill, alcohol or certain medicines, unaccustomed exercise, and combining EMS with other hard training the same week.
- Recognise the warning signs: muscle pain far beyond normal soreness, swelling, weakness, and **dark or cola-coloured urine** — and know these need urgent medical attention.
- Prepare the client: a light carbohydrate snack about 2 hours before; 250–500 ml of water (about 5 ml per kg body weight) 30 minutes before and again straight after the session; no alcohol, drugs or muscle relaxants for 24–48 hours before; no session with a fever.
- Apply Hasta's recovery rules and the conditioning phase (module 6), and give the post-session advice sheet.

Escalation — no exceptions

Dark urine, severe pain or swelling after a session: advise the client to get medical attention the same day, inform the clinician, record an incident in the app, and book no further sessions until the clinician clears the client.

MODULE 4 Machine module (per model)

Supplier's course, at least 4 hours · practical

Aim: safe, correct use of the specific machine. A trainer is certified only for the machine models they have completed this module on.

Covers: the supplier's official course; device and suit inspection before use; suit sizing and fit; electrode positions and moistening as the supplier specifies; controls, programmes and per-area intensity; emergency stop — within reach of both trainer and client; cleaning, drying and storage;

recognising faults; when a device must be taken out of use and marked "under maintenance" in the app.

Assessed: set-up, fitting, emergency-stop demonstration and cleaning, observed by a Master Trainer qualified on the model.

MODULE 5 Running a session

4 hours · practical

Aim: every session is run the same safe way, from greeting to notes.

The session, step by step:

1. **Before:** admission complete and in date; pre-session questions (Part C checklist); at least the minimum recovery gap since the last session; hydrated and has eaten.
2. **Set-up:** inspect the device and suit; fit suit and electrodes; check placement avoids wounds, implants and restricted areas; explain what the client will feel and how to signal "less" or "stop"; show the emergency stop.
3. **Intensity:** raise each electrode area gradually while asking the client using the Borg CR10 scale (Part C). Target the level set for the client's stage — first sessions start around **4 ("somewhat strong")**. Never "10" (maximal), never continuous cramp-like contraction, never pain.
4. **During:** one client per trainer at Hasta; stay with the client and keep talking; check each area at least 3 times during the session and adjust; correct movements by voice and touch; stop at once if anything feels wrong.
5. **After:** cool-down; ask how they feel; hydration and recovery advice; book the next session no sooner than the recovery rule allows; clean and store the suit.
6. **Record:** session notes, intensities used, client feedback and any concerns in the app the same day.

Movement during stimulation: simple, controlled exercises the client can do safely (for example partial squats, lunges, rows, chest presses, trunk work) timed with the impulse phase; no jumping or ballistic movements in early sessions.

MODULE 6 Programme design and progression

3 hours · theory + case study

Aim: plan a client's sessions across their package so they progress safely.

By the end, the trainee can:

- Apply the **conditioning phase** for new clients (Part C): the international guideline advises no high-intensity or exhausting sessions in the first 8–10 weeks, about one 20-minute session a week, a first session of about 5 minutes continuous familiarisation then about 12 minutes of intermittent work, and very hard intensities (CR10 7–8) only after 8–10 sessions at the earliest.
- After conditioning, keep at least 4 days between intense sessions (the guideline's rule), or Hasta's rule if the medical lead sets a stricter one.
- Plan Hasta packages (for example 10 or 20 sessions) week by week, and explain the plan and its timescale to the client honestly.
- Adapt for goals (strength and tone, back health as cleared by the clinician, support for weight management alongside food and activity), and for older or less active clients.
- Recognise when to hold or step back: unusual soreness, poor sleep, illness, missed weeks.

DECISION **Gap between sessions**

Hasta currently books EMS sessions at least **48 hours** apart. The international guideline is more cautious for beginners (about one session a week for 8–10 weeks) and asks for at least 4 days between intense sessions afterwards. The medical lead should decide Hasta's rule before approving this curriculum; the EMS app can then enforce it when sessions are booked.

MODULE 7 Emergencies, adverse events and first aid

2 hours · scenario (BLS certificate required separately)

- Emergency stop: where it is, who can reach it, and using it without hesitation.
- Fainting or dizziness: stop, lie the client down with legs raised, monitor, get help.
- Cramp, unusual pain, skin burning or redness under an electrode: stop, check, record.
- Chest pain, breathlessness, palpitations, confusion: stop, call for medical help.
- Unresponsive and not breathing normally: call emergency services (112) and start basic life support.
- Reporting: every adverse event or near miss is recorded in the incident log in the app the same day and reviewed by the medical lead.

Activities: timed emergency-stop drill; simulated fainting and adverse-reaction scenarios; completing an incident report.

MODULE 8 Measurements, progress and lifestyle support

2 hours · practical

- Take and record measurements consistently in the app (weight, girth sites, body composition where available, photos only with consent and only on clinic devices).
- Read progress over time and explain it honestly — including when results are slow.
- Give general advice within a trainer's scope: water, regular meals with protein, sleep, daily activity.
- Know the limits: diets, supplements, medicines and medical conditions go to the clinic's doctor or dietitian. Refer, don't prescribe.

MODULE 9 Hasta protocols, the platform and data protection

2 hours · theory

- Admission, consent and the rule that no session runs without a complete admission.
- Packages: what each includes, entitlements, expiry, and that payments are taken in the clinic app against the package bill.
- Booking rules, rotas, device availability and maintenance status.
- Session notes, measurements and incident reports in the EMS app.
- Confidentiality: health information is shared only within the care team; no patient photos or details on personal phones, WhatsApp or social media (see the confidentiality agreement and the clinic phone policy, office order HCPL/OO/2026/001); obligations under the Digital Personal Data Protection Act, 2023.

MODULE 10 Client care, ethics and professional conduct

1 hour · theory

- Respect, privacy and dignity: changing area, suit fitting, chaperone on request.
- Honest expectations — no promises of outcomes, no pressure selling.
- Working within your scope and asking when unsure.
- Handling complaints and concerns, and escalating them to the manager.

MODULE 11 Supervised practice

At least 10 sessions · log + sign-off

The trainee runs real client sessions with a Master Trainer qualified on the machine present throughout. Each session is logged in the EMS app with the Master Trainer's sign-off. At least 3 must be clients' first EMS sessions. The practical checklist (Part C) is assessed during these sessions; every critical item (★) must be passed.

Part C — Tools

C1. Suggested timetable

Day	Morning	Afternoon
Day 1	Module 1 — How EMS works (3 h)	Module 2 — Screening, contraindications and consent (part 1, 2 h)
Day 2	Module 2 (part 2, 2 h) · Module 3 — Rhabdomyolysis (part 1, 1 h)	Module 3 (part 2, 2 h) · Module 7 — Emergencies (2 h)
Day 3	Module 4 — Supplier machine course	Module 4 (continued)
Day 4	Module 5 — Running a session (4 h)	Module 6 — Programme design (3 h)
Day 5	Module 8 (2 h) · Module 9 (2 h)	Module 10 (1 h) · revision · theory test in the app
Weeks 2–6	Module 11 — supervised sessions and practical checklist, then certification in the app	

C2. Pre-session checklist (every session)

	Check
☐	Admission complete and consent in date (renewed at least every 6 months)
☐	Any change since the last session: illness or fever, new medicine, pregnancy, surgery, injury, new implant?
☐	Alcohol, drugs or muscle relaxants in the last 24–48 hours?
☐	How did they feel after the last session — unusual soreness, swelling, dark urine?
☐	Recovery gap since the last session meets Hasta's rule
☐	Eaten a light snack; drunk 250–500 ml of water in the last 30 minutes
☐	Device, cables and suit inspected; emergency stop within reach of both of us
☐	No electrode over wounds, rashes, implants or areas the clinician has restricted

C3. Borg CR10 scale — how hard does it feel?

CR10	Description	Use at Hasta
0–2	Nothing to weak	Familiarisation only
3	Moderate	Warm-up
4	Somewhat strong	**First sessions** (conditioning phase start)
5–6	Strong	Gradual progression during the conditioning phase
7–8	Very strong / very hard	Only after 8–10 sessions at the earliest, and only if the client tolerates it well
9–10	Extremely strong / maximal	**Never**

C4. Conditioning phase for a new client (international guideline)

When	Session
Session 1	About 5 minutes continuous low-intensity familiarisation, then about 12 minutes intermittent (about 4 s impulse with 0.3–0.5 s ramps, about 4 s rest). Target CR10 about 4.
Weeks 1–3	Up to 20 minutes, CR10 4–5, no exhausting sessions.
Weeks 4–10	Up to 20 minutes intermittent (strength-type) or 30–40 minutes continuous (endurance-type, moderate). Progress CR10 gradually; 7–8 only after 8–10 sessions.
After conditioning	At least 4 days between intense sessions (CR10 ≥ 7).

Frequency during conditioning: the guideline advises about one 20-minute session a week. Hasta's rule — see the DECISION in module 6.

C5. Practical assessment checklist (★ = critical)

	The trainee…
★	Confirms the admission is complete and checks for changes since the last session (health, medicines, pregnancy, illness)
★	Checks the recovery gap and asks about soreness or dark urine after the last session
	Checks hydration and food; advises water before and after
	Inspects the device, cables, suit and electrodes before use
	Fits the suit and electrodes correctly; moistens contact areas as the supplier specifies
★	Avoids placement over wounds, rashes, implants or restricted areas
	Explains what the client will feel and how to signal discomfort
	Selects the programme approved for the client's stage and the clinician's precautions
★	Increases intensity gradually per muscle group, within first-session limits
	Watches and talks with the client throughout; adjusts intensity on feedback
★	Demonstrates the emergency stop and responds correctly to a simulated adverse reaction
	Completes cool-down and checks how the client feels before they leave
	Gives recovery and hydration advice and books the next session within the recovery rule
	Cleans and stores the suit and electrodes as the supplier specifies
	Records session notes, concerns and incidents in the platform
	Maintains privacy, dignity and professional conduct throughout

C6. Decisions for the medical lead before approval

#	Decision	Current Hasta rule
1	Minimum gap between sessions in the conditioning phase (guideline: about 1 session a week for 8–10 weeks)	48 hours
2	Minimum gap between intense sessions after conditioning (guideline: 4 days)	48 hours
3	First-session limits: duration, programme and maximum CR10	"Set by the medical lead" — not yet written down
4	Final list of absolute and relative contraindications for Hasta (module 2)	Admission screening in the EMS app
5	Approve the question bank, including the new questions added with this version	All questions are still drafts

Approved by the medical lead (name, registration no., date)

Approved in the EMS app — Academy → Curriculum v1.1

References: Kemmler W, et al. Position statement and updated international guideline for safe and effective whole-body electromyostimulation training — the need for common sense in WB-EMS application. Front Physiol. 2023;14:1174103. · von Stengel S, et al. Revised contraindications for the use of non-medical WB-electromyostimulation. Evidence-based German consensus recommendations. Front Sports Act Living. 2024;6:1371723.